SHUN DONG

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University of Kansas, School of Business
Lawrence, KS

EDUCATION

University of Kansas — Lawrence, KS
Ph.D. Candidate in Analytics, Information and Operations GPA: 3.84 | Expected May 2027

University of Cincinnati — Cincinnati, OH
M.S. in Statistics GPA: 3.72/4.0

Missouri State University — Springfield, MO
M.S. in Mathematics, Statistics focus GPA: 3.82/4.0
M.B.A. GPA: 3.91/4.0
B.S. in Mathematics, Actuarial Science and Statistics focus GPA: 3.48/4.0

RESEARCH INTERESTS

Statistical Learning and Machine Learning; Business Analytics and Information Systems; Reliable AI and Model Calibration; Optimization and Decision Support; Applications in Finance and Cybersecurity.

JOB MARKET PAPER

Accounting for Industry Heterogeneity in Predicting Corporate Default Probability: A Semiparametric Learning Approach [JOB MARKET PAPER]

with Shaobo Li, Ben Sherwood, and Xiaorui Zhu

Revise and Resubmit, European Journal of Operational Research

Abstract: Predicting corporate default is essential for risk management, credit allocation, and investment decision making, yet most bankruptcy models overlook heterogeneity in how financial distress manifests across industries. We propose the single-index hazard model with multiple links (SIML), a semiparametric learning method that combines a common financial risk index with industry-specific flexible link functions. This structure captures nonlinear industry effects and their interactions with firm-level financial characteristics while retaining an interpretable low-dimensional index. We also develop a two-stage simulation-based testing procedure for industry heterogeneity and nonlinearity. Using a comprehensive database of 18,651 U.S. publicly traded firms from 1980 to 2023, we find strong evidence of distinct nonlinear risk patterns across industries. SIML outperforms a broad set of benchmark models in both in-sample and out-of-sample evaluations. A simulated corporate loan market and an asset-pricing analysis of the distress risk anomaly further show the economic value of accounting for industry-specific default risk.

PUBLICATIONS

Relationship between piperacillin concentrations, clinical factors and piperacillin/tazobactam-associated acute kidney injury

Tang Girdwood, S., Hasson, D., Caldwell, J. T., Slagle, C., **Dong, S.**, Fei, L., Tang, P., Vinks, A. A., Kaplan, J., & Goldstein, S. L.
Journal of Antimicrobial Chemotherapy, 78(2), 478-487 (2023). DOI: 10.1093/jac/dkac416.

Diagnostic accuracy of serum matrix metalloproteinase-7 as a biomarker of biliary atresia in a large North American cohort

Pandurangi, S., Mourya, R., Nalluri, S., Fei, L., **Dong, S.**, Harpavat, S., Guthery, S. L., Molleston, J. P., Rosenthal, P., Sokol, R. J., Wang, K. S., Ng, V., Alonso, E. M., Hsu, E. K., Karpen, S. J., Loomes, K. M., Magee, J. C., Shneider, B. L., Horslen, S. P., Teckman, J. H., & Bezerra, J. A.
Hepatology (2024). DOI: 10.1097/HEP.0000000000000827.

Maternal metabolic dysfunction-associated steatotic liver disease during pregnancy and its impact on offspring steatosis

Klepper, C. M., Trout, A. T., **Dong, S.**, Fei, L., Stankiewicz, T. E., Wayland, J., Moreno-Fernandez, M. E., Divanovic, S., Xanthakos, S. A., DeFranco, E., Woo, J. G., & Mouzaki, M.
Clinical Nutrition ESPEN, 70, 327-335 (2025). DOI: 10.1016/j.clnesp.2025.10.009.

WORKING PAPERS

Accounting for Industry Heterogeneity in Predicting Corporate Default Probability: A Semiparametric Learning Approach [JOB MARKET PAPER]

with Shaobo Li, Ben Sherwood, and Xiaorui Zhu

Revise and Resubmit, European Journal of Operational Research

Cost Restricted Feature Selection with Information Criterion

with Ben Sherwood and Shaobo Li

Accepted for presentation at the 2026 INFORMS Workshop on Data Science

Preparing for submission to Management Science

Robust Outcome Optimization via Multi-Model Inverse Classification

with Michael Lash, Qihang Lin, and Nick Street

Target journal: Journal of Machine Learning Research

Objective-Aligned Sparse Learning for Neyman-Pearson Classification: Applications to Corporate Default and Cyber Threat Detection

Manuscript in preparation

Dynamic Instrumented Principal Components for Time-Varying Risk Exposures

Manuscript in preparation

TEACHING EXPERIENCE

University of Kansas, School of Business, Lawrence, KS 2024 to Present

TA / Instructor, Database Management Systems (BSAN 326 / IST 326 / IST 328)

Independently taught multiple undergraduate sections in Fall 2024, Fall 2025, and Fall 2026.

Instructor evaluation means ranged from **4.10 to 4.36 out of 5** across three evaluated sections.

University of Kansas, School of Business, Lawrence, KS 2022 to Present

TA, Advanced Machine Learning (BSAN 745)

TA, Data Mining and Predictive Analytics (BSAN 450)

University of Cincinnati, Department of Mathematics, Cincinnati, OH 2018 to 2020

TA / Instructor, Elementary Statistics I (STAT 1034) and Elementary Statistics II (STAT 1035)

TA / Recitation Instructor, Calculus I (MATH 1061) and Calculus II (MATH 1062)

Missouri State University, Department of Mathematics, Springfield, MO 2015 to 2017

TA / Lab Instructor, MTH 101 Intermediate Algebra A

TA, Calculus I & II (including occasional classroom instruction)

PRESENTATIONS

"Corporate Probability of Default: A Single-Index Model with Multiple-Link Approach"

2026 INFORMS Annual Meeting - Job Market Showcase Track
POMS Annual Conference
Big XII + MIS Research Symposium

2024 Joint Statistical Meetings (JSM)

2023 CMStatistics
INFORMS Annual Meeting

"Robust Outcome Optimization via Multi-Model Inverse Classification"

2025 INFORMS Annual Meeting

"Cost Restricted Feature Selection with Information Criterion"

2026 INFORMS Workshop on Data Science

SERVICE

Faculty Assembly Representative, KU School of Business 2026–Present
Represent the interests of Ph.D. students in faculty assembly meetings and participate in school-level governance, discussions, and voting.

Session Chair,
Production and Operations Management Society (POMS) Annual Conference (2026)
Chaired research presentations and facilitated discussion and Q&A among participants.

Ad Hoc Reviewer 2024-Present
Journal of Business Analytics (2026)
INFORMS Workshop on Data Science (2026);
INFORMS Data Mining Best Paper Competition (2026);
International Conference on Information Systems (ICIS) (2025);
International Workshop on Health Intelligence (2024)

HONORS AND AWARDS

School of Business Best Paper Award, University of Kansas	2026
Max E. Fessler Dissertation Support Award, University of Kansas	2026
John O. Tollefson Best Paper Award, University of Kansas	2026
Graduate Teaching Assistantship, University of Kansas	2023-Present
Graduate Research Assistantship, University of Cincinnati	2020-2022
Graduate Teaching Assistantship, University of Cincinnati	2018-2020
Graduate Teaching Assistantship, Missouri State University	2015-2017
1st Place, CNAS Thesis Competition Research Award, Missouri State University	2014
CSSA Outstanding Contribution Award	2014

RESEARCH AND PROFESSIONAL EXPERIENCE

Cincinnati Children's Hospital Medical Center, Cincinnati, OH 2020 to 2022
Graduate Research Intern.
Collaborated with biostatisticians and physicians on statistical and bioinformatics research, applying advanced statistical methods to investigate disease and chronic-condition risk factors.

China Continent Insurance LLC, Linyi City, Shandong Province, China Summer 2013 and Summer 2014
Manager Assistant.
Assessed vehicle condition for depreciation. Followed up with the insured individuals regarding premium and deductibles payments.

ADDITIONAL INFORMATION

Programming and Computing: R (Rcpp, RcppArmadillo), Python, C, C++, SQL, SAS, MATLAB, LaTeX, Microsoft Office; **Other Experience and Certification:** Insurance Agent Qualification Certificate, 2013

REFERENCES

Shaobo Li
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University of Kansas School of Business |
shaobo.li@ku.edu

Ben Sherwood
Associate Professor; Jack and Shirley Howard Mid-Career Professor
University of Kansas School of Business |
ben.sherwood@ku.edu

Debabrata "Deb" Dey
Davis Area Director, Analytics, Information, Operations; Ronald G. Harper Professor of Artificial Intelligence and Information Systems
University of Kansas School of Business |
deb.dey@ku.edu

Lin Fei
Professor, UC Department of Pediatrics
Cincinnati Children's Hospital Medical Center |
lin.fei@cchmc.org